UNIVERSITY OF ICELAND

Wind Speed and Rural Traffic Accidents in Iceland

A Frequency-Adjusted Analysis, 2007–2025

Ásgeir Snær Magnússon

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M.Sc. thesis
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FACULTY OF INDUSTRIAL ENGINEERING, MECHANICAL ENGINEERING AND COMPUTER SCIENCE

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30 ECTS thesis submitted in partial fulfillment of a
Master of Science degree in Software Engineering

Supervisors
Kristján Jónasson
Guðmundur Freyr Úlfarsson

Faculty of Industrial Engineering, Mechanical Engineering and Computer Science
School of Engineering and Natural Sciences
University of Iceland
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Faculty of Industrial Engineering, Mechanical Engineering and Computer Science
School of Engineering and Natural Sciences
University of Iceland
Dunhagi 5
107, Reykjavik, Iceland
Telephone: 525 4000

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Abstract

This thesis examines whether strong mean wind speeds are disproportionately common when rural injury accidents occur in Iceland. Accident records, cleaned 10-minute weather observations, urban boundaries, annual road-section traffic statistics, and daily counter data are combined in a documented analysis. The primary analysis includes 6,192 rural injury accidents from 2007–2025 with valid wind observations within 20 km. Within each weather station, calendar year, and season, observed accidents in each mean-wind interval are compared with the number expected from the local background frequency of that interval. Confidence intervals are obtained from 5,000 bootstrap samples clustered by weather station.

The observed/expected ratio is close to one across most low and moderate wind intervals. At mean wind speeds of 20–25 m/s, 61 accidents were observed compared with 29.0 expected, giving a ratio of 2.11 (95% station-clustered bootstrap interval 1.54–2.71). Maximum gust is a secondary wind measure. The result describes relative accident occurrence, not a causal effect or accident risk per vehicle. Traffic data are unavailable at the required spatial and temporal resolution for the full study period. The thesis documents data cleaning, matching, coverage and statistical analysis separately.

Útdráttur

Ritgerðin rannsakar hvort hár meðalvindhraði sé óvenju algengur þegar umferðarslys með meiðslum verða í dreifbýli á Íslandi. Hún sameinar slysaskrá, hreinsuð tíu mínútna veðurgögn, þéttbýlismörk, árleg umferðargögn fyrir vegkafla og daglegar umferðartalningar. Aðalgreiningin nær til 6.192 slysa með meiðslum á árunum 2007–2025 sem tengjast gildri vindmælingu innan 20 km. Fyrir hverja veðurstöð, hvert ár og hverja árstíð er fjöldi slysa í vindbili borinn saman við þann fjölda sem vænta mætti miðað við staðbundna vindtíðni.

Við meðalvind upp á 20–25 m/s urðu 61 slys, en 29,0 voru vænt. Hlutfall séðra og væntra slysa var því 2,11 (95% bil 1,54–2,71). Vindhviður eru skoðaðar sem aukagreining. Niðurstaðan lýsir hlutfallslegri tíðni slysa, en ekki orsakaáhrifum eða áhættu á hvert ökutæki. Umferðargögn eru ekki tiltæk með nægilega nákvæmri staðsetningu og tímasetningu fyrir allt rannsóknartímabilið.

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Abbreviations

ADU	Average annual daily traffic.
SDU	Summer daily traffic, June–September.
VDU	Winter daily traffic, December–March.
VHDU	Spring/autumn daily traffic derived from ADU, SDU and VDU.
O/E	Observed accidents divided by expected accidents.

1. Introduction

Strong wind can affect vehicle stability, lane keeping, driver behaviour, and the decision to travel. Iceland is a useful setting for studying this question because wind conditions vary strongly across locations and time. However, accident counts alone are not accident risk. A severe weather event can raise the risk faced by each vehicle while simultaneously reducing the number of vehicles on the road. Weather frequency and traffic exposure must therefore be treated as separate parts of the problem [Andrey et al.(2003), Lord and Mannering(2010), Washington et al.(2020)].

The primary analysis focuses on mean wind speed and rural accidents involving injury. Maximum gust and traffic results are reported separately.

1.1. Objective and Research Questions

The objective is to estimate how rural injury-accident occurrence differs by mean-wind interval after accounting for how frequently each interval occurs locally. Maximum gust is examined using the same method as a secondary measure.

1. By how much does the observed/expected ratio for rural injury accidents differ across 5 m/s mean-wind intervals, relative to local background mean-wind frequency?
2. By how much does the corresponding observed/expected ratio differ across maximum-gust intervals?

The primary result estimates relative accident occurrence. It does not estimate a causal effect or a national accident rate per vehicle-kilometre.

The analysis records weather-cleaning rules, keeps unmatched accidents in the matching data, reports spatial coverage, calculates local background wind frequency, and estimates uncertainty by weather-station-clustered bootstrap. The figures and tables can be regenerated from documented scripts [Peng(2011)].

2. Data and Analysis Files

2.1. Source Websites and Downloaded Data

The source websites and the downloaded content are listed explicitly here so that the origin of every analytical variable can be audited. Accident records were supplied as tab-separated exports from Samgöngustofa; its public accident register description and statistics page are available at `https://www.samgongustofa.is/umferd/tolfraedi/umferdarslys/` [Samgöngustofa(2026)]. Ten-minute observations and station metadata were obtained from the Icelandic Meteorological Office data service at `https://api.vedur.is/weather/` [Veðurstofa Íslands(2026)]. Annual traffic workbooks and daily counter reports were downloaded from the Vegagerðin traffic pages at `https://www.vegagerdin.is/samgongukerfid/vegakerfid/umferdartolur` and `https://umferd.vegagerdin.is/` [Vegagerðin(2026)]. Urban polygons were downloaded from the Statistics Iceland WFS endpoint `https://gis.is/geoserver/Hagstofan/ows`. Current counter locations and road-section geometry were obtained from the Vegagerðin ArcGIS layers `https://vegasja.vegagerdin.is/arcgis/rest/services/data/`.

Table 2.1.: Source websites and downloaded datasets. The source identifiers are used in Table 2.3. The primary study period is 2007–2025.

ID	Website	Downloaded dataset	Records and period	Fields used
ACC	Samgöngustofa accident register	Accident events and road links	126,607 events, 2007–2025; 126,610 road links, 2007–2025	`nid`, date, time, coordinates, `meidsli`, `tegohapps`, and registered road section.
IMO	Icelandic Meteorological Office API	Ten-minute weather and station metadata	226,580,952 observations, 2007–2025; 771 station-history records	station, time, mean wind `f`, maximum gust `fg`, and station coordinates.
IRCA-A	Vegagerðin traffic workbooks	Annual road-section traffic	33,757 section-years, 2000–2025	year, road section, length, `ADU`, `SDU`, and `VDU`.
IRCA-D	Vegagerðin counter reports	Daily counter channels	1,033,659 channel-days, 2019–2024	date, counter identifier, road section, reported station, direction/lane channel, and count.
IRCA-R	Vegagerðin MapServer	Counter locations and road geometry	Current register and network reference	counter identifier, coordinates, road section, station range, and geometry.
SI	Statistics Iceland WFS	Urban-area polygons	97 polygons, 2020–2024 publication	polygon geometry and locality identifiers used for rural classification.

The 2025 accident, injury, vehicle, road-link, annual-traffic, and weather records pass through the same preparation rules as the earlier years and are included in the primary analysis. Daily counter reports currently end in 2024.

Table 2.2 shows the first chronological records in the accident-event source file. It illustrates that the original delivery contains source codes and Icelandic field names; the analysis keeps the original injury code rather than replacing it with a derived severity label. The displayed `flokkur2` code is part of the unmodified source extract but is not carried into the analytical files.

Table 2.2.: First chronological records in the accident-event source file. Location text is shortened only for display.

nid	Date	Time	x	y	meidsli	tegohapps	stadsetn	flokkur2
3166374	01.01.2007	00:38:00	361264	405675	3	121	Á Sæbraut, skammt norðan við Miklubrautarbrúna	4
3166372	01.01.2007	00:44:00	358064	406363	4	920	Við gatnamót á Bústaðavegi, Select	6
3166394	01.01.2007	01:14:00	359475	405515	4	142	Á Bústaðavegi, LHS, gegnt	2
3166406	01.01.2007	01:27:00	355877	400981	4	270	Við Miðvang, hús nr. 123	3
3166596	01.01.2007	04:42:00	356779	408494	4	920	Við Garðastræti 6	6
3166700	01.01.2007	06:31:00	356995	407945	4	710	Við Skálholtsstíg 2	9
3166730	01.01.2007	06:58:00	358845	405767	4	510	Gatnamót Bústaðavegar og Brúarinnar	4
3166883	01.01.2007	13:44:00	277239	496763	4	920	Gatnamót Ólafsbrautar og Ennisbrautar	6
3166887	01.01.2007	13:48:00	401235	384366	3	12	S-vegur, Fossnes 14	1
3166897	01.01.2007	14:17:00	484011	561541	4	21	Sfj.-vegur, Syðsta-Grund	1

Table 2.3.: Processing pipeline and analysis files. The compact files in `data/analysis/` *are the routine inspection and core-analysis inputs.*

Step	Inputs	Output file	Records and period	Purpose and fields used
Accident preparation and match	ACC, SI, IMO	`analysis/accidents.csv`	6,414 events, 2007–2025	Rural injury events with `nid`, time, `meidsli`, `tegohapps`, vehicle count, matched station, distance, time difference, `f`, and `fg`.
Wind cleaning	IMO	`processed/weather/weather\_10min\_clean.parquet`	211,497,897 observations, 2007–2025	Applies the fixed wind-quality rules before any frequency calculation. Fields: station, time, `f`, and `fg`.
Wind-frequency table	Clean weather, IMO station metadata	`analysis/weather\_frequency.csv`	646,209 station-year-season-bin rows, 2007–2025	Counts each wind bin and its denominator using station, year, season, variable, bin, and measurement count.
Annual traffic	IRCA-A	`analysis/annual\_traffic.csv`	33,757 section-years, 2000–2025	Standardises road section, length, ADU, SDU, and VDU.
Daily counter traffic	IRCA-D, IRCA-R, clean weather	`analysis/daily\_traffic.csv`	774,274 counter-days, 2019–2024	Daily count, road section, location method, matched station and distance, daytime `f` and `fg`.
Primary O/E	Accident and frequency files	`mean\_wind\_oe.csv`	6 wind bins, 2007–2025	Observed and expected accidents by 5 m/s mean-wind bin, standardised within station, year, and season.
Stratified crash-rate ratio	Road-period cache, shared-station accidents, and daily traffic factors	`stratified\_crash\_rate\_ratio\_by\_wind.csv`	6 wind bins; 4,958 accidents, 2007–2025	Compares estimated accident rates within road section, year, and traffic period using vehicle-kilometres as an offset.

2.2. Accident Data

The accident source covers 2007–2025 [Samgöngustofa(2026)]. The study population comprises 6,414 rural accidents involving injury. Accident severity is based on the source field `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The primary outcome is therefore `meidsli` ≤ 3. The serious-or-fatal comparison uses `meidsli` ≤ 2. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons [Hagstofa Íslands(2026)]. Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are therefore not represented.

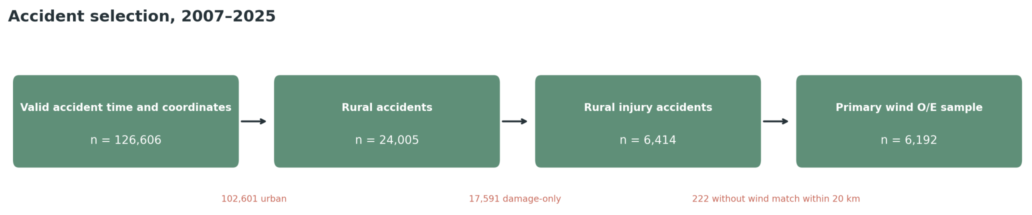

Figure 2.1.: Available accident records, severity of the rural injury sample, and coverage of the primary weather match. Damage-only accidents remain available but are not part of the injury outcome.

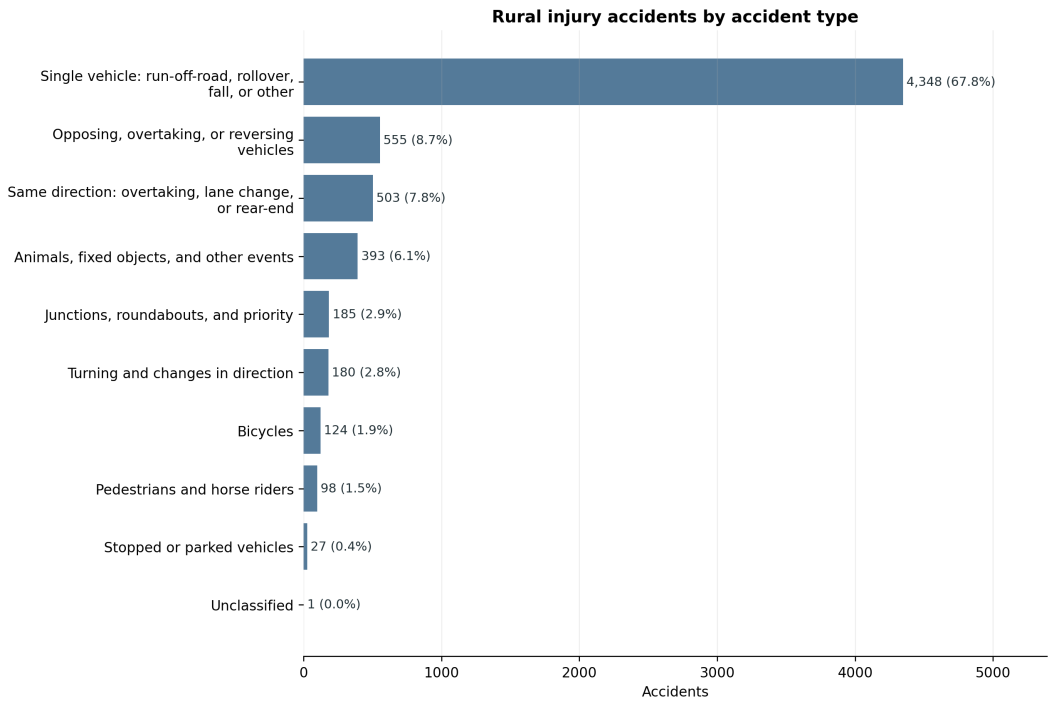

Figure 2.2.: Accident-type distribution among the 6,414 rural injury accidents.

2.3. Weather Data

The weather source contains 10-minute observations from Icelandic weather stations. The variables used here are mean wind speed (`f`) and maximum wind gust (`fg`). Temperature (`t`) remains in the weather file but is not required for the wind analysis. A wind observation is used when

$$0 \leq f < 45, \qquad 0 \leq fg < 65, \qquad fg + 0.5 \geq f.$$

Rows missing either `f` or `fg` are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where `fg = 0` but `f > 0` are excluded and reported in the annual audit. A value of `f = 0` alone is retained. Only uninterrupted all-zero runs where `f = fg = 0` for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a clean wind observation at either adjacent timestamp on the 10-minute observation grid. The observations therefore have 10-minute resolution, but the selected observation is no more than five minutes from the recorded accident time. The primary maximum distance is 20 km. The matching file retains all 6,414 accidents and records whether each accident meets the weather-match criterion.

Table 2.4.: Wind-quality audit among records from station-years with wind data, 2007–2025

Category	Records	Share of assessed records
Records assessed for wind quality	214,770,209	100.00%
Missing `f` or `fg`	427,778	0.20%
Negative or upper-threshold wind	300,649	0.14%
Internally inconsistent `f`/`fg`	2,268	<0.01%
Frozen all-zero runs ($\geq$ 2 hours)	2,541,617	1.18%
Clean wind observations retained	211,497,897	98.48%

The detailed audit reports each cleaning rule separately. The audit and the weather-distribution figure use the full 2007–2025 weather delivery. The O/E analysis uses only observations matched to accidents from 2007–2025.

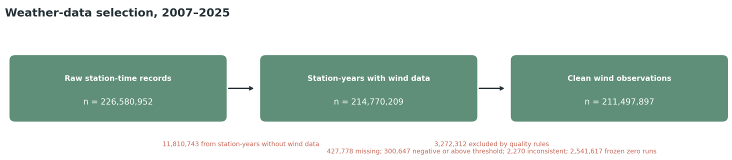

Figure 2.3.: Weather observations retained after the documented wind-quality rules.

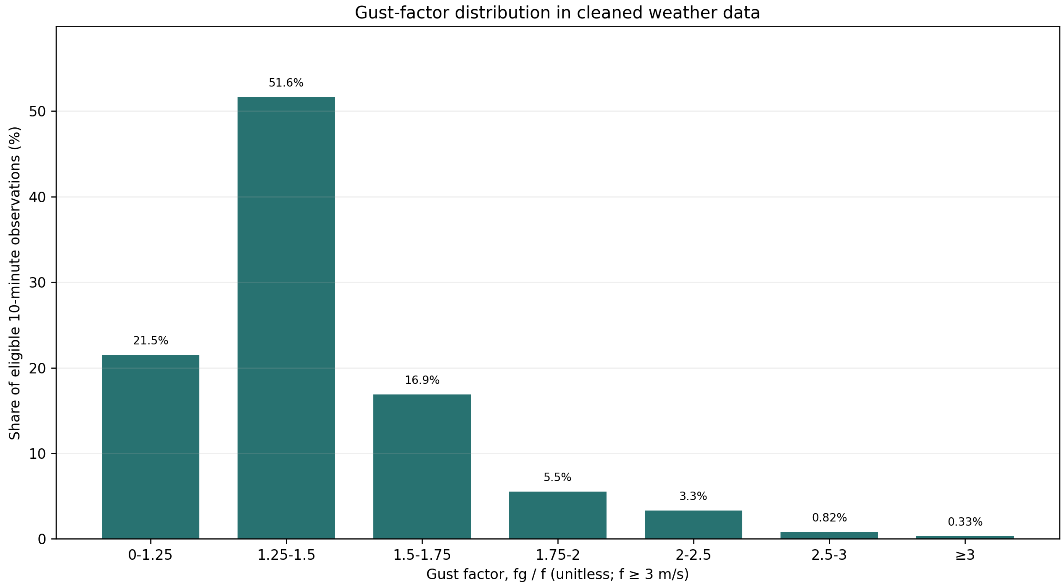

Figure 2.4.: Distribution of gust factor (fg/f) in the cleaned weather observations. The ratio is calculated only where the 10-minute mean wind speed is at least 3 m/s, so that values from near-calm conditions do not dominate the distribution. Each eligible 10-minute observation has equal weight.

2.4. Traffic Data

Two traffic sources are used for different purposes.

Annual road-section files provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 [Vegagerðin(2026)]. The road-section comparison uses the years shared with the accident study, 2007–2025. Following the official definitions, SDU covers June–September and VDU covers December–March. These files can be linked to accidents with a registered road-section code, but exact matching is much narrower than the weather matching.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024. They describe observed traffic at counter sites and are useful for measuring travel demand during windy days. They do not provide hourly vehicle-kilometres at every accident location. More than one counter may also occur on a road section. Each report identifies a counter by road section and *stöð* (road station). The annual road files provide the corresponding section start and end stations (`Bst` and `Est`), which permit a counter position to be interpolated along the section geometry when an official counter coordinate is not available.

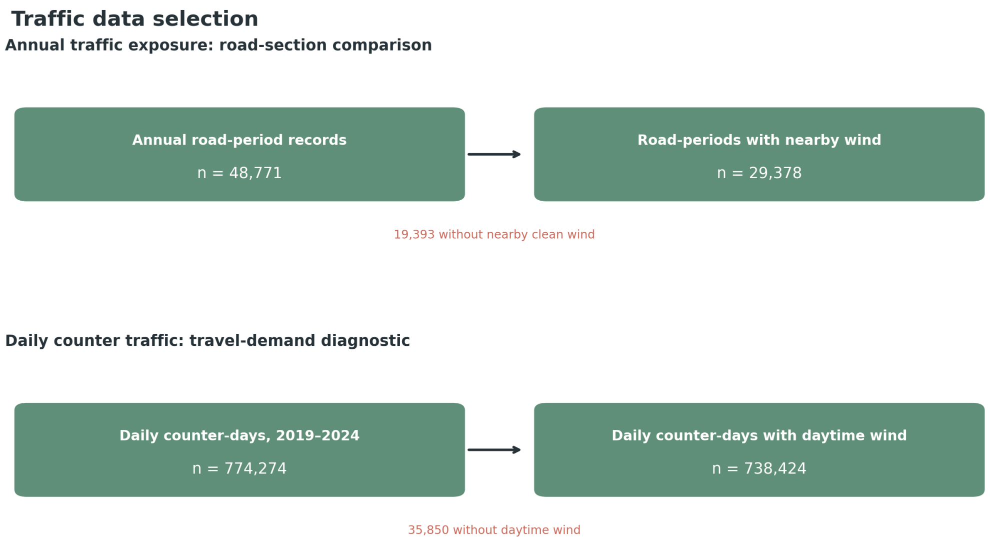

Figure 2.5.: Coverage of the annual road-section data, the road-section wind table, and the separate daily-counter diagnostic.

2.5. Reproducible Implementation

The repository groups the code into weather, accident, traffic, and analysis modules. The weather module cleans observations, the accident module performs the weather match, and the analysis module produces O/E estimates, bootstrap intervals, road-section results, and the daily counter diagnostic. The repository README identifies the individual programs. The reported results can be regenerated with

```
.venv/bin/python -m src.analyze
```

Figures are generated by the analysis scripts and are not manually edited.

3. Methods

3.1. Primary Wind-Frequency Analysis

Mean wind speed `f` is the pre-specified primary exposure and is divided into 5 m/s intervals. Maximum gust `fg` is retained as a supporting analysis. Each accident is compared with the mean-wind distribution at its matched weather station in the same calendar year and season. The expected count in a mean-wind interval is therefore based on how often that interval occurred locally, rather than on a national wind average.

Let g identify a weather station, calendar year, and meteorological season. Let A_{jg} be the number of accidents in mean-wind interval j and group g, and let $A_g = \sum_j A_{jg}$. From all clean 10-minute observations, let T_{jg} be the number in interval j and T_g the total in group g. The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the mean-wind interval. A value above one means that accidents are over-represented in that interval. Standardisation within station, year, and season prevents windy stations or seasons from dominating solely because their wind distributions differ.

Uncertainty is estimated with 5,000 weather-station-clustered bootstrap replicates. Stations are sampled with replacement, while all years, seasons, weather observations, and accidents associated with a sampled station remain together. The reported interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

3.2. Why Traffic Is Not in the Primary Ratio

The primary denominator is background wind time. Traffic is not included because it is unavailable at the spatial and temporal resolution required for the full study period. The estimated crash-rate analysis uses annual traffic on every eligible road section, rather than only sections where an accident occurred. It includes 4,958 accidents that have an exact annual road-section, year, and traffic-period link. It uses SDU and VDU where they are published, and a clearly labelled residual VHDU estimate for April–May and October–November.

Using those traffic data in the primary curve would change the study population and mix seasonal average traffic with 10-minute accident-time wind. The primary curve therefore measures *wind-frequency-adjusted relative accident occurrence*. It does not itself measure accident rate per vehicle.

3.3. Road-Section Traffic Analysis

Traffic is included where the spatial and temporal keys can be linked explicitly. For road section, year, official traffic period, and wind interval k, j, estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where q_k is SDU, VDU, or derived VHDU vehicles per day, L_k is section length, D_k is the number of calendar days, and p_{kj} is the local share of clean 10-minute observations in the wind interval. The accident rate is

$$\text{Rate}_j = \frac{A_j}{\sum_k V_{kj}} \times 100{,}000{,}000.$$

This method accounts for road length and for spatial and seasonal differences in published traffic. It still assumes that traffic within a traffic period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute. The accident wind is measured at the nearest qualifying station to the accident location; the wind frequency used to allocate traffic is measured at the nearest qualifying station to the road-section midpoint. For this analysis, the accident wind is re-matched to that same station and retained only when the station is also within 20 km of the accident.

The principal traffic result is a conditional Poisson model. Each road section, year, and traffic period has its own intercept. For stratum k and wind interval j, the model is

$$\log E(A_{kj}) = \alpha_k + \beta_j + \log V_{kj},$$

where A_{kj} is the accident count and $\log V_{kj}$ is the offset. The exponentiated coefficient $\exp(\beta_j)$ is the estimated rate ratio relative to 0–5 m/s within otherwise identical road-section–year–period strata.

The daily counter analysis provides a separate, counter-informed allocation. For wind interval j, its traffic factor is the observed-to-typical daily traffic ratio in that interval, divided by the corresponding ratio at 0–5 m/s. Within each road-section–year–period stratum, V_{kj} is multiplied by this factor and then rescaled so that the stratum's total annual vehicle-kilometres are unchanged. Thus, it changes only the assumed allocation of traffic across wind intervals; it is not a direct observation of vehicle-kilometres at each 10-minute weather measurement.

3.4. Observed Daily Traffic Diagnostic

Daily traffic is analysed separately because its weather definition is daily. Directional channels are first summed to one 24-hour count for each physical counter and date. The unit is then one fixed counter on one date; separate counter sites on the same road section are not summed. For each counter, year, month, and weekday, expected traffic is the arithmetic mean of the available daily counts in that same group. The traffic index is

$$I_{cd} = 100 \times \frac{Q_{cd}}{\mathrm{mean}(Q_{c,\,\text{same year, month, weekday}})}.$$

An index below 100 indicates fewer counted vehicles than usual for that local calendar context. For each wind interval, the observed-to-expected traffic ratio is aggregated across counter-days and expressed relative to the 0–5 m/s ratio. This is the factor used in the counter-informed allocation above. It describes the participating counter sites rather than national traffic. The 24-hour traffic count is not converted into hourly traffic. Mean wind is calculated from clean observations between 10:00 and 21:59, so this is a daily association rather than an estimate of traffic in that window. Official counter coordinates are used where a conservative location link is available. Otherwise the report's *stöð* is placed proportionally between the year-specific `Bst` and `Est` on the registered road-section geometry and is marked as estimated. The nearest station within 20 km having usable observations that day is then selected. This interpretation of *stöð* is independently checked against 105 conservative name-matched official counter locations: the median position difference is 2.6 m (90th percentile 179 m). A road-section midpoint is retained only where the station range or geometry is unavailable. Uncertainty in the median index is estimated by resampling whole counters 1,000 times.

For the small residual set of midpoint-based locations, the table records the maximum possible offset if the counter were anywhere on the registered section. These are conservative location-uncertainty bounds, not estimates of the counter's actual error. Station-interpolated locations do not receive this midpoint bound: their uncertainty is historical and geometric rather than a known interval around a midpoint.

Counter-year mean traffic is compared with exact road-section ADU as a consistency check. A near-complete counter-year has at least 300 observed days; single-counter road sections are evaluated separately because multiple PDF records can represent different locations, directions, or lanes.

4. Results

4.1. Coverage

Table 4.1.: Coverage of the retained analyses

Analysis step	Available	Scope	Coverage
Wind match within 10 km	4,858	6,414	75.74%
Wind match within 20 km (primary)	6,192	6,414	96.54%
Wind match within 30 km	6,399	6,414	99.77%
Exact annual road-section match, 2007–2025	5,731	6,414	89.35%
Stratified crash-rate analysis	4,958	6,192	80.07%
Daily counter-days with daytime wind	738,424	774,274	95.37%

The 20 km threshold retains most accidents while limiting the spatial distance to the weather proxy. Increasing the threshold to 30 km adds 207 accidents but can be misleading in complex terrain. The road-section analysis is substantially narrower than the primary analysis and must not be presented as equally representative.

The two annual-traffic rows have different analytical meanings. The first records an exact road-section and year link. The second also requires a defined traffic period and usable annual traffic and weather-frequency exposure. It is the population used for the estimated crash-rate analysis below.

4.2. Mean Wind Speed and Relative Accident Frequency

Ratios are close to one across most low and moderate intervals. The plotted ratios increase from 15 m/s. At mean wind speeds of 20–25 m/s, 61 accidents were observed compared with 29.0 expected. The O/E ratio is 2.11, with a station-clustered interval of 1.54–2.71. At mean wind speeds of at least 25 m/s, 16 accidents were observed compared with 5.9 expected (O/E 2.70; 95% interval 0.97–5.49). The upper tail contains few accidents, so the results describe a broad pattern rather than precise thresholds.

The same accidents, weather records, matching rule, and O/E calculation can also be grouped into 3 m/s intervals. That presentation is included in Appendix A.4. The wider 5 m/s grouping is used in the main figure because it gives a direct, readable description of the high-wind pattern without implying that the bin boundaries are physical thresholds.

Maximum gust is a secondary measure. Its O/E pattern rises more strongly in the highest gust intervals; at gusts of at least 36 m/s, 25 accidents were observed compared with 4.2 expected (O/E 5.99; 95% interval 3.62–8.80). This result is reported as supporting evidence, not as a replacement for the mean-wind primary analysis.

Table 4.2.: Highest-gust result under three station-distance rules

Maximum distance	Matched accidents	Observed at ≥ 36	O/E	95% interval
10 km	4,858	25	7.14	4.35–10.43
20 km (primary)	6,192	25	5.99	3.62–8.80
30 km	6,399	26	5.89	3.54–8.68

4.3. Maximum Gust as a Supporting Analysis

Maximum gust is examined with the same station, year, and season standardisation. The gust O/E figure is provided in Appendix A.2.

4.4. Estimated Accident Rates per Vehicle-Kilometre

The stratified model includes 4,958 accidents in 4,004 informative road-section–year–period strata. Relative to 0–5 m/s, the estimated rate ratio is 1.59 (95% CI 1.36–1.86) at 15–20 m/s, 2.36 (1.76–3.16) at 20–25 m/s, and 4.52 (2.77–7.37) at at least 25 m/s. The time-proportional allocation assumes that traffic within a period follows local wind frequency.

Daily counters show lower traffic on windier days. Relative to 0–5 m/s, the daily traffic factors are 0.86 at 15–20 m/s, 0.75 at 20–25 m/s, and 0.62 at at least 25 m/s. Applying these factors before allocating annual vehicle-kilometres raises the corresponding rate ratios to 1.85, 3.16, and 7.09. This second allocation is a supporting comparison: its wind measure is a daytime daily mean, whereas the rate model uses 10-minute wind observations.

4.5. Descriptive Subgroups

Appendix figures repeat the mean-wind and gust calculations by season and by the number of vehicles involved. They are descriptive checks, not separate research

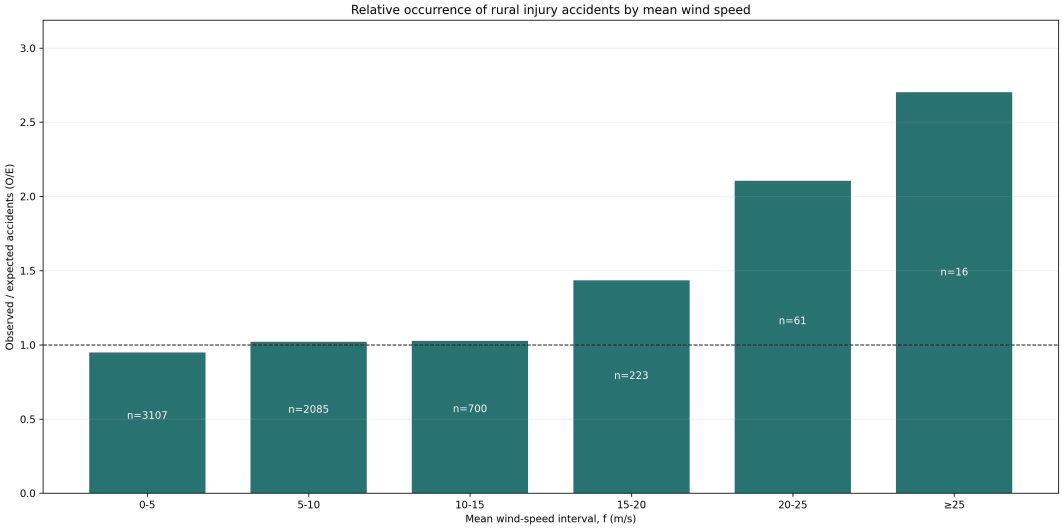

Figure 4.1.: Wind-frequency-adjusted relative occurrence of rural injury accidents by mean wind speed. The label inside each bar is the observed accident count. Traffic volume is not included. Bootstrap intervals are reported in the supporting results rather than drawn on the figure.

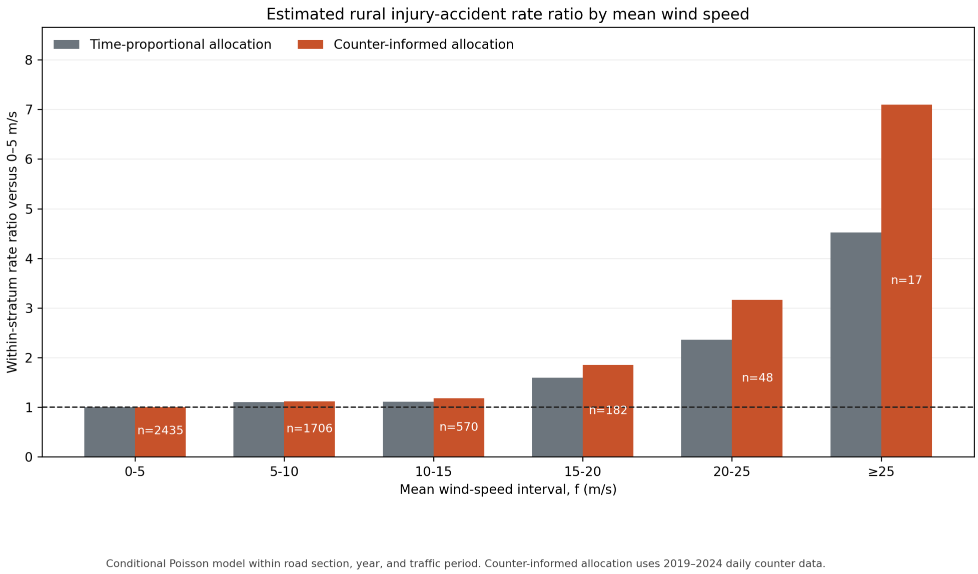

Figure 4.2.: Estimated rural injury-accident rate ratios relative to 0–5 m/s. The model compares wind intervals within the same road section, year, and traffic period. Grey bars allocate seasonal traffic in proportion to local 10-minute wind frequency. Orange bars use the observed daily-counter traffic pattern as an additional allocation factor. Labels show observed accidents.

questions or tests of interaction; the high-wind cells become sparse after the sample is divided. Daylight and road-surface subgroups are not presented. A valid O/E subgroup requires the same classification for accident times and for every background weather observation. Daylight has not yet been derived for the complete station-time denominator, and the available road- surface history describes road construction rather than the wet, icy, or dry condition at the accident time. Presenting either comparison now would be misleading.

4.6. Daily Traffic During Wind

The daily counter dataset contains 1,033,659 directional daily records. Directional channels at the same physical counter site are summed, while separate counter sites on the same road section remain separate. This produces 774,274 counter-days from 476 sites. Usable daytime wind is available for 738,424 counter-days (95.37%). Traffic remains close to its local typical level below 9 m/s and declines at higher mean wind speeds. This pattern is consistent with lower travel demand during strong wind. It does not estimate national traffic or traffic at the time of each accident. At mean daytime wind speeds of at least 24 m/s, the observed count is 62.6% of the expected count for a typical day at the same counter, year, month, and weekday. This upper interval contains 294 counter-days at 105 counters, so it is supporting descriptive evidence rather than a replacement for the primary O/E denominator.

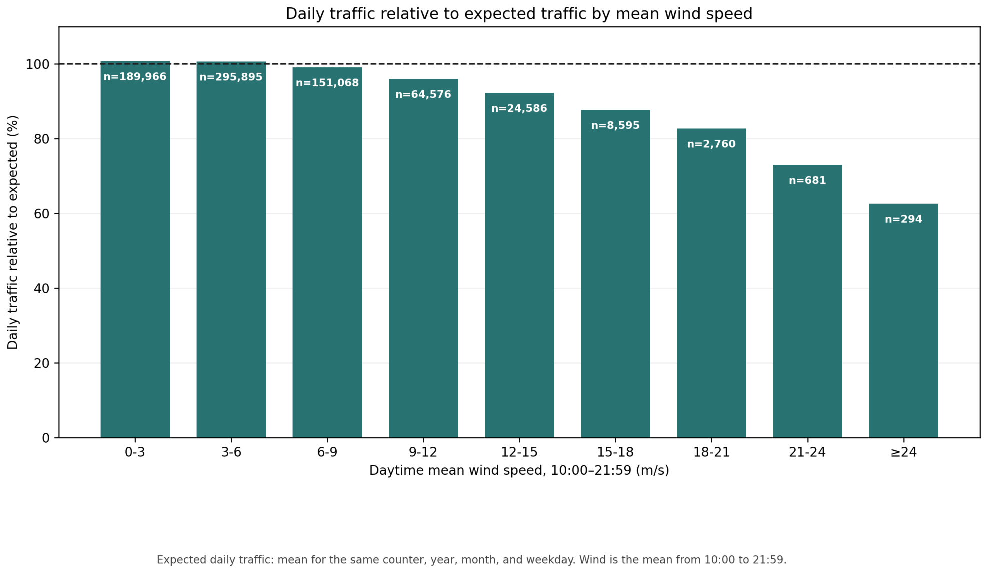

Figure 4.3.: Vehicles counted per day relative to a typical day at the same counter, year, month, and weekday. The labels show the number of counter-days. Daytime mean wind is calculated from 10:00 to 21:59.

5. Discussion

5.1. Interpretation

Strong mean wind is over-represented at the times of rural injury accidents relative to its local frequency. The comparison is standardised within weather station, calendar year, and season, and uncertainty is clustered by station. The result describes association rather than a per-vehicle causal effect. Moderate wind intervals fluctuate around one, while several upper intervals contain few accidents.

Traffic remains important for interpretation. The primary curve cannot separate risk per vehicle from the number of vehicles exposed. In the traffic analysis, rate ratios increase after comparison within road section, year, and traffic period. Daily counters show lower traffic during strong wind, and the counter-informed allocation strengthens the high-wind pattern. These results make it less likely that the main O/E pattern is explained by increased travel during severe wind. The available traffic data are still too limited for a directly observed national per-vehicle rate.

5.2. Strengths

The main strengths are the large 2007–2025 accident sample, 10-minute weather resolution, explicit spatial coverage reporting, local wind-frequency standardisation, and station-clustered bootstrap intervals. The matching data retain unmatched accidents and record the reason for each unsuccessful match. The limited scope keeps the connection between the research question, method, and result clear.

5.3. Limitations

- The primary result is not traffic-adjusted. It should not be interpreted as accident risk per vehicle-kilometre.
- A weather station up to 20 km away is a proxy for conditions at the accident location. Terrain can produce important differences over shorter distances.
- The stratified crash-rate analysis represents 4,958 accidents and uses annual seasonal traffic allocated by wind frequency, rather than traffic observed at the accident minute. It aligns accident weather and exposure to the same station,

but excludes accidents for which that station is farther than 20 km from the accident.

- The counter-informed allocation uses 2019–2024 daily traffic and daytime daily mean wind. It supports the direction of the adjustment, but it is not a direct measurement of vehicle-kilometres in each 10-minute wind bin.
- Daily traffic counts cover selected counter sites. Multiple counters can represent different directions, lanes, or locations; retaining counters separately avoids double counting but does not provide national coverage.
- The highest wind intervals contain relatively few accidents. Several intervals and comparison results are shown, so individual estimates in the upper tail should be interpreted with caution.
- Fixed 2020–2024 urban boundaries are applied to the entire 2007–2025 accident period.
- The analysis is observational. Unmeasured road condition, vehicle type, speed, visibility, and driver behaviour can contribute to the pattern.

5.4. Next Improvements

The most valuable additional data would be verified hourly traffic counts linked to stable road-section identifiers, section geometry, and the same weather stations used in the accident analysis. That would allow accident-time gust and non-accident vehicle exposure to be measured at compatible temporal resolution. A future analysis could use wider high-wind intervals, vehicle class, and exposed-road geometry, and could apply a hierarchical count model once adequate traffic exposure is available. Better exposure data would add more value than additional models based on the current information.

6. Conclusion

This thesis examines wind speed and rural injury accidents in Iceland. Of 6,414 accidents from 2007–2025, 6,192 were matched to a valid wind observation within 20 km. After standardisation for local mean-wind frequency within station, year, and season, accidents were over-represented at mean wind speeds of 20–25 m/s: 61 were observed compared with 29.0 expected, giving an O/E ratio of 2.11 with a 95% station-clustered interval of 1.54–2.71. The secondary gust analysis shows a stronger upper-tail pattern.

Daily counter traffic falls during strong wind, and the stratified vehicle-kilometre rate ratio also rises at higher mean wind speeds. Neither traffic source provides complete hourly exposure for the full study period. The evidence therefore shows an association between strong wind and higher relative occurrence of rural injury accidents. Better linked hourly traffic data are required to estimate directly observed road risk per vehicle.

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A. Supporting Analyses

A.1. Accident Characteristics

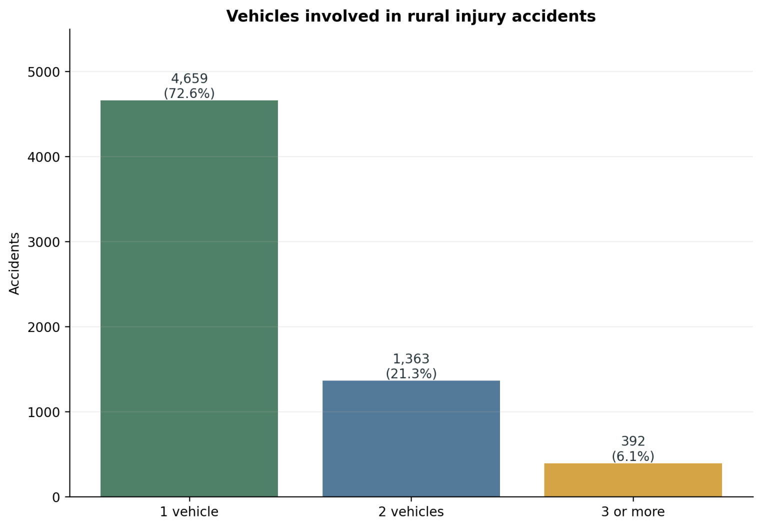

Figure A.1.: Number of registered vehicles involved in each rural injury accident.

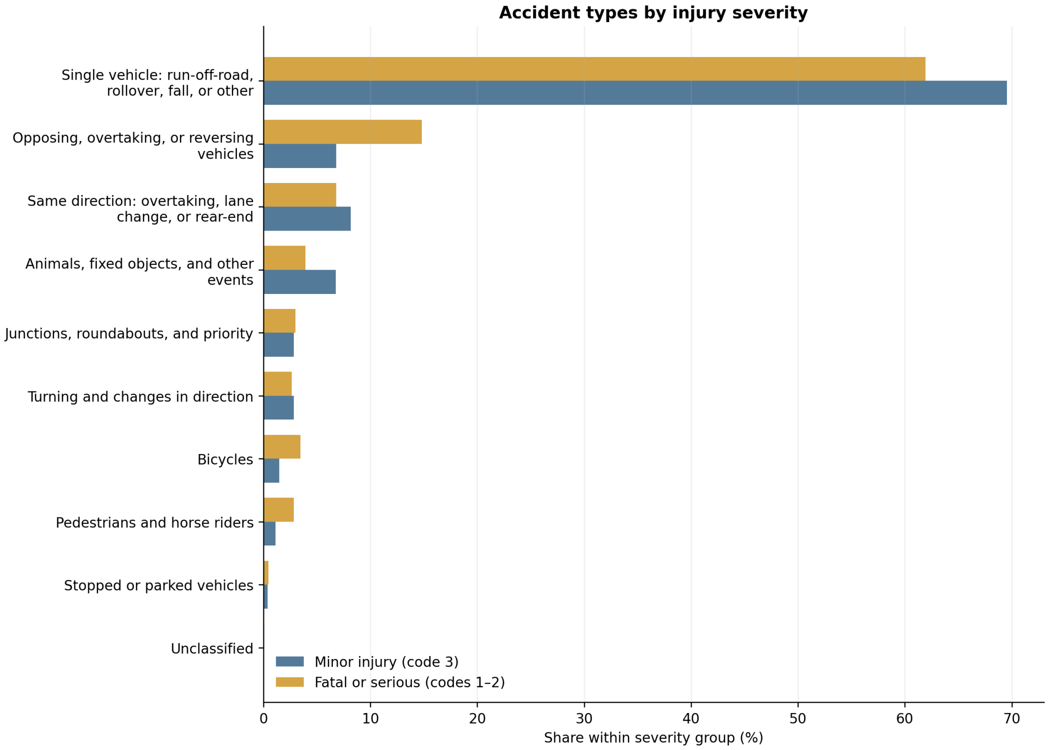

Figure A.2.: Accident-type composition in minor-injury accidents and in the non-overlapping fatal-or-serious group. Percentages sum to 100 within each severity group.

A.2. Maximum Gust O/E

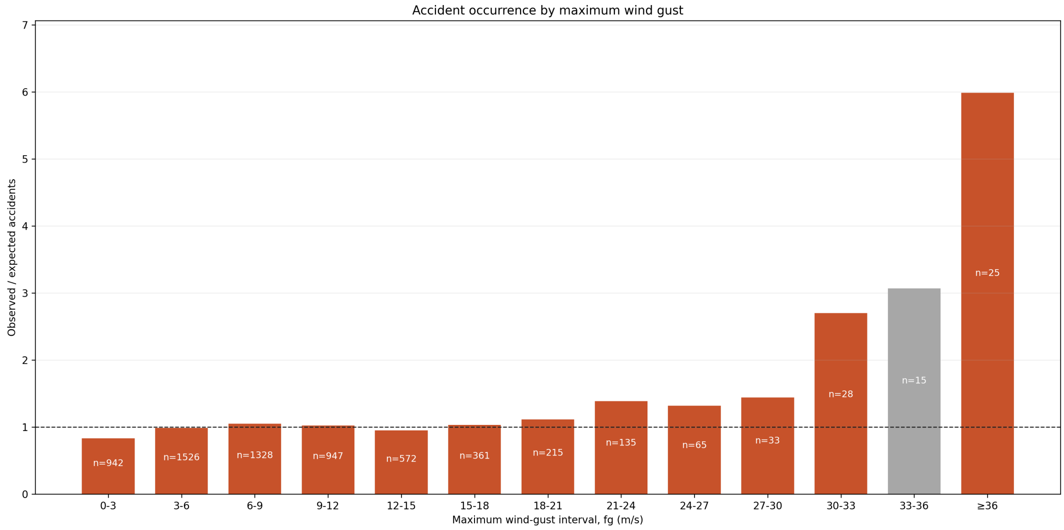

Figure A.3.: Frequency-adjusted observed/expected accident occurrence by maximum wind gust. This is a supporting analysis; mean wind speed is the primary exposure.

A.3. Gust Factor O/E

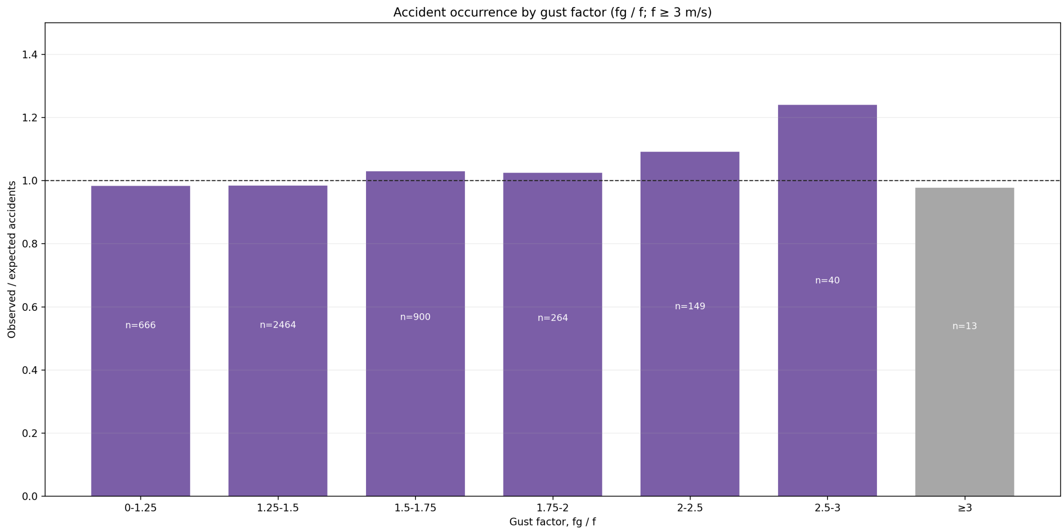

Figure A.4.: Frequency-adjusted observed/expected accident occurrence by gust factor (fg/f). The ratio is calculated only where mean wind speed is at least 3 m/s. This exploratory analysis does not show a clear monotonic pattern.

A.4. Mean Wind Grouped in 3 m/s Intervals

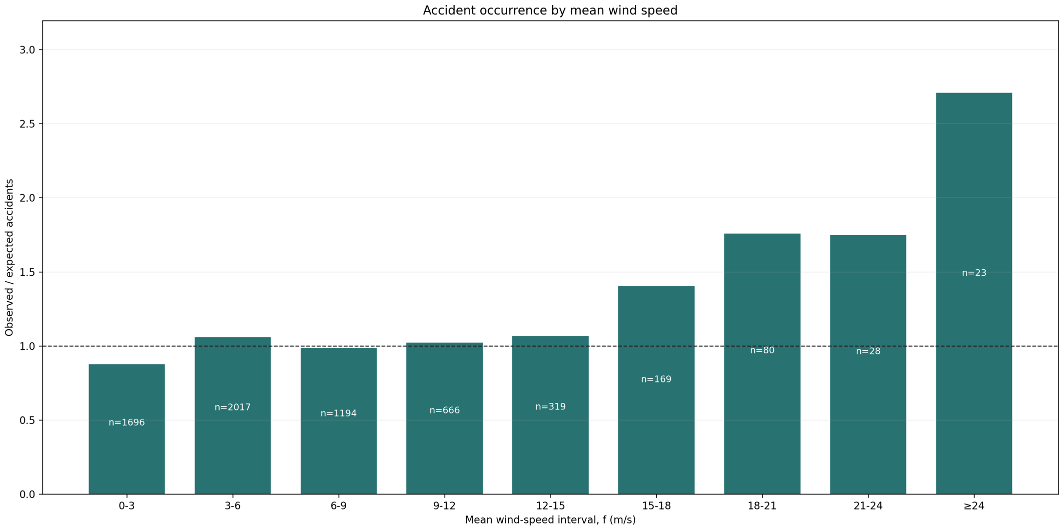

Figure A.5.: The primary mean-wind O/E calculation presented in narrower 3 m/s intervals. The accident sample, weather records, matching rule, and standardisation are unchanged; only the grouping shown on the horizontal axis is different.

A.5. Mean-Wind Subgroups

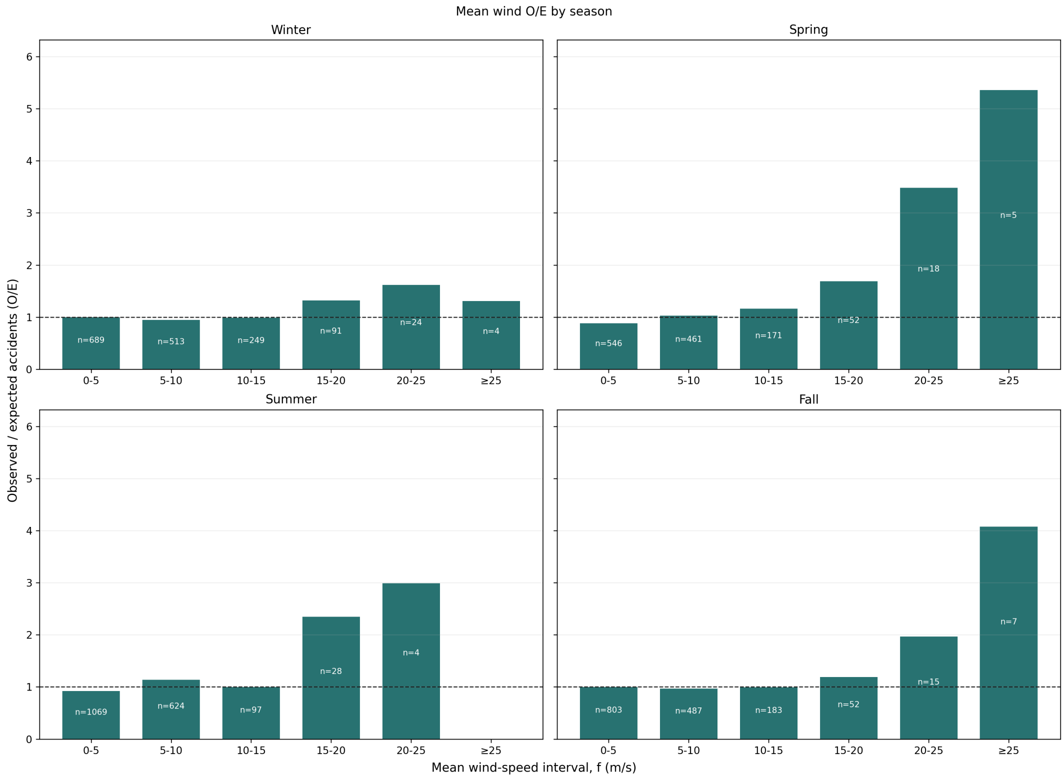

Figure A.6.: Mean-wind O/E estimates by meteorological season. These estimates are shown as a descriptive subgroup analysis; high-wind bins are sparse after the sample is divided by season.

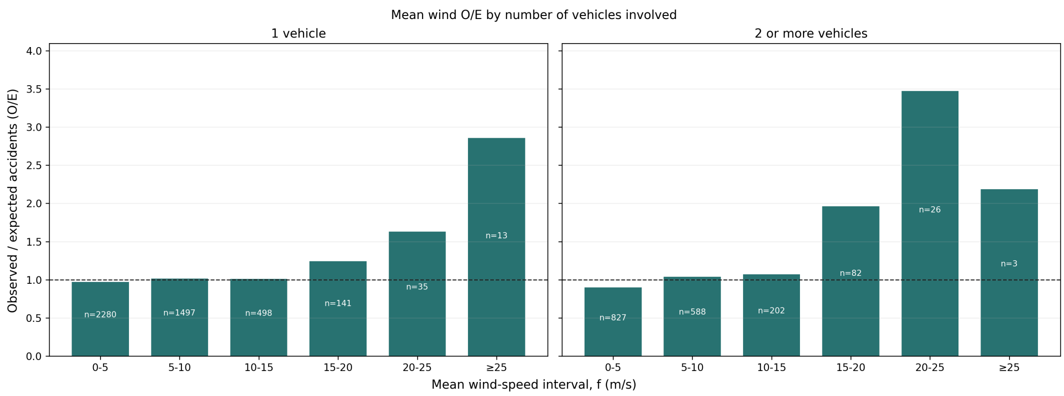

Figure A.7.: Mean-wind O/E estimates by the number of registered vehicles involved. This is a descriptive subgroup analysis; it does not establish accident type or vehicle count as an explanatory variable.

A.6. Maximum-Gust Subgroups

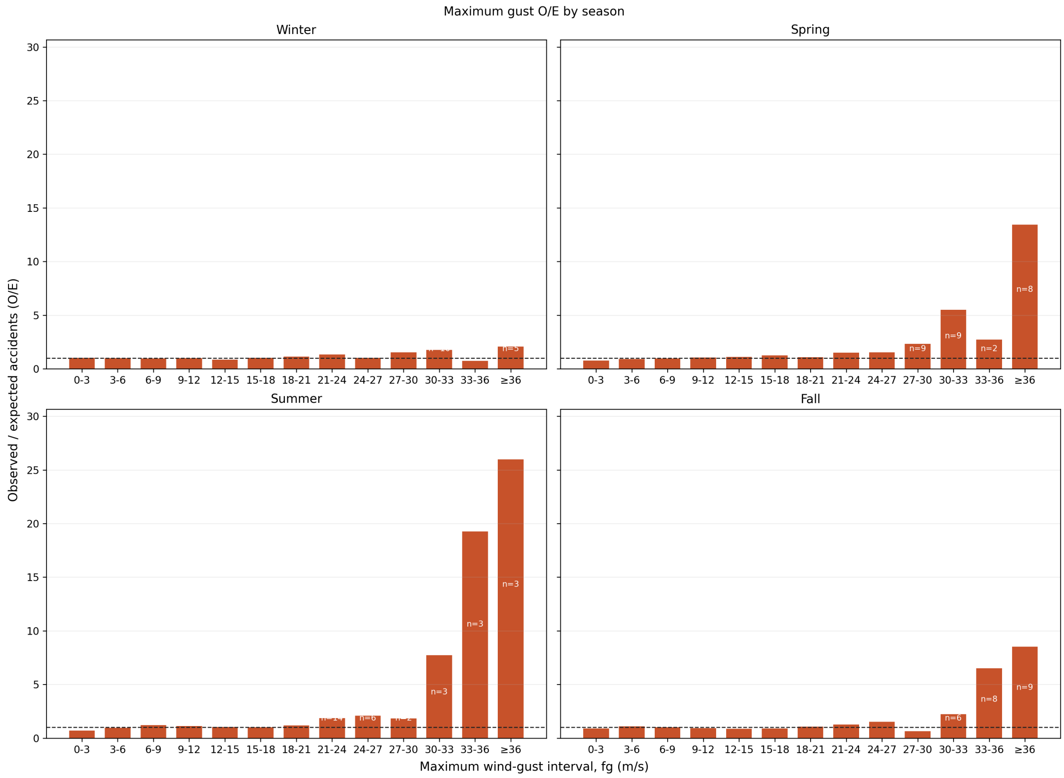

Figure A.8.: Maximum-gust O/E estimates by meteorological season. The high-gust cells are sparse after the sample is divided, so the panels are descriptive.

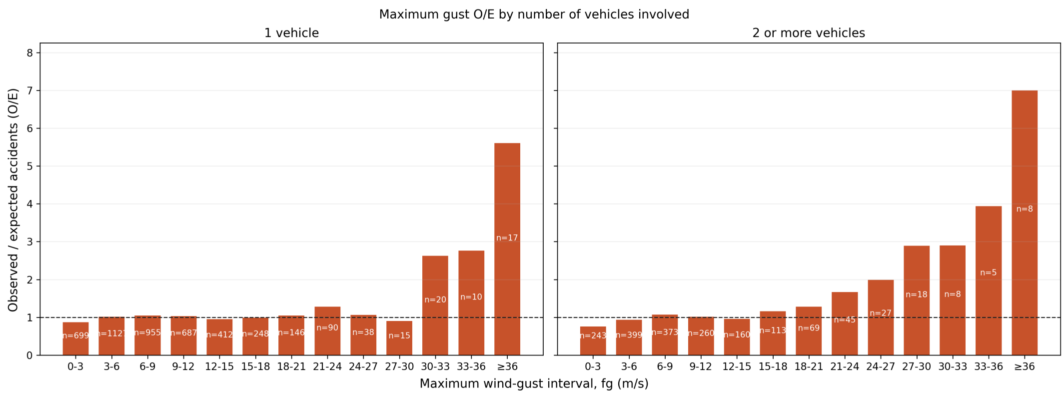

Figure A.9.: Maximum-gust O/E estimates by the number of registered vehicles involved. This is a descriptive subgroup analysis.

A.7. Daily-Counter Validation

Table A.1.: Consistency of observed daily PDF traffic with official ADU

Scope	N	Median	P10–P90	$r_{\log}$
All exact matches	2,328	0.975	0.245–1.115	0.942
At least 300 observed days	1,906	0.984	0.240–1.063	0.940
At least 300 days, one counter per section	1,122	1.000	0.925–1.112	0.993

A.8. Stratified Vehicle-Kilometre Model

Table A.2 gives the intervals omitted from the bar chart. The time-proportional allocation is the principal traffic result. The counter-informed allocation is a supporting comparison because it uses daily daytime mean wind from selected counters in 2019–2024.

Table A.2.: Conditional Poisson rate ratios relative to 0–5 m/s

Mean wind	Accidents	Time-proportional ratio (95% CI)	Counter-informed ratio (95% CI)
0–5	2,435	1.00 (reference)	1.00 (reference)
5–10	1,706	1.10 (1.03–1.17)	1.12 (1.05–1.19)
10–15	570	1.11 (1.01–1.22)	1.18 (1.08–1.30)
15–20	182	1.59 (1.36–1.86)	1.85 (1.58–2.16)
20–25	48	2.36 (1.76–3.16)	3.16 (2.36–4.23)
$\geq$ 25	17	4.52 (2.77–7.37)	7.09 (4.33–11.63)

The vehicle-group and seasonal versions were also estimated. They are not shown as figures because the upper intervals are too sparse for a balanced comparison: there are three two-or-more-vehicle accidents, four winter VDU accidents, and three summer SDU accidents at mean wind speeds of at least 25 m/s. At 15–20 m/s, the estimated time-proportional ratios are 1.32 (1.08–1.61) for one-vehicle accidents and 2.31 (1.79–2.97) for accidents with two or more vehicles. The corresponding winter VDU and summer SDU ratios are 1.43 (1.15–1.78) and 2.30 (1.63–3.23). These are descriptive subgroup results, not separate main findings.